Zhuoyue Wan

Ph.D. Candidate in Computer Science

Department of Computing
The Hong Kong Polytechnic University
Hung Hom, Kowloon, Hong Kong

Homepage: wanzhuoyue.com Email: zhuoy.wan@connect.polyu.hk

EDUCATION

Jan. 2025 - Present Ph.D. in Computer Science, **The Hong Kong Polytechnic University**
Supervisor: Prof. Chen Jason Zhang
Research collaborator: Prof. Raymond Chi-Wing Wong

Sep. 2022 - Nov. 2023 M.S. in Data-Driven Modeling, **The Hong Kong University of Science and Technology**

Sep. 2017 - Jun. 2021 B.S. in Statistics, **Chongqing University**

RESEARCH EXPERIENCE

Jan. 2024 - Dec. 2024 **Research Assistant**, The Hong Kong Polytechnic University

RESEARCH INTERESTS

Natural Language Interfaces to Data: Text-to-SQL, Text-to-Visualization, and Text-to-OverpassQL.

Data Preparation for LLM: Quality and diversity assessment of pre-training data, noisy data cleaning, and coreset selection.

Data Agent & Optimization: Autonomous data agents and mathematical optimization for performance and efficiency.

PUBLICATIONS | PUBLISHED & ACCEPTED

- **Zhuoyue Wan**, Jiaqi Fan, Hwanhee Kim, Chen Jason Zhang, Shuaimin Li, Yuanfeng Song, Haoyang Li, Xiao-Yong Wei, Raymond Chi-Wing Wong. XOver: Execution-Calibrated Exemplar Contextualization for Multilingual Text-to-OverpassQL on OpenStreetMap. **ICDE 2027, to appear.** (CCF-A)
- **Zhuoyue Wan**, Wentao Hu, Hwanhee Kim, Chen Jason Zhang, Shuaimin Li, Yuanfeng Song, Ming Deng, Xiao-Yong Wei, Raymond Chi-Wing Wong. TACO: Token-Aware Context Optimization for Structured OpenStreetMap Query Generation. **SIGMOD 2027, to appear.** (CCF-A)
- **Zhuoyue Wan**, Wentao Hu, Chen Jason Zhang, Yuanfeng Song, Shuaimin Li, Ruiqiang Xiao, Xiao-Yong Wei, Raymond Chi-Wing Wong. OsmT: Bridging OpenStreetMap Queries and Natural Language with Open-source Tag-aware Language Models. **ICDE 2026.** (CCF-A)
- **Zhuoyue Wan**, Yuanfeng Song, Shuaimin Li, Chen Jason Zhang, Raymond Chi-Wing Wong. DataVisT5: A Pre-trained Language Model for Jointly Understanding Text and Data Visualization. **ICDE 2025.** (CCF-A)
- Jiaqi Fan, **Zhuoyue Wan**, Chen Jason Zhang, Xiao-Yong Wei, Haoyang Li, Yuanfeng Song, Wangze Ni, Raymond Chi-Wing Wong. Text-to-RLS: Towards Constraint-Anchored Row-Level Security in Natural Language Database Interfaces. **ICDE 2027, to appear.** (CCF-A)
- Wentao Hu, **Zhuoyue Wan**, Jinhao Shen, Chen Jason Zhang, Xiaoyong Wei, Qing Li. Meta-Moderator: Empowering Multi-Agent Debate with Meta-Cognition. **Findings of EMNLP 2026.**
- Shuaimin Li, Liyang Fan, Zeyang Li, **Zhuoyue Wan**, Yufang Lin, Shiwen Ni, Feiteng Fang, Hamid Alinejad-Rokny, Yuanfeng Song, Kun Jing, Chen Jason Zhang, Min Yang. SrDetection: A Self-Referential Framework for Data Leakage Detection in Code Large Language Models. **Findings of ACL 2026.**
- Shuaimin Li, Yuanfeng Song, Xuanang Chen, Anni Peng, **Zhuoyue Wan**, Chen Jason Zhang, Raymond Chi-Wing Wong. VisPoison: An Effective Backdoor Attack Framework for Tabular Data Visualization Models. **ICDE 2026.** (CCF-A)
- Luyang Luo, Xin Huang, Minghao Wang, **Zhuoyue Wan**, Hao Chen. Medical Image Debiasing by Learning Adaptive Agreement from a Biased Council. **Medical Image Analysis, 2025.** (JCR-Q1)

PREPRINTS

- Huahang Li, Wentao Hu, **Zhuoyue Wan**, Chen Jason Zhang, Haoyang Li, Xiaoyong Wei. DataClaw: An Autonomous Data Agent with Instant Messaging Integration. **arXiv preprint, 2026.**
- Yan Luo, **Zhuoyue Wan**, Nemin Wu, Yuzhong Chen, Gengchen Mai, Fu-lai Chung, Kent Larson. TransFlower: An Explainable Transformer-Based Model with Flow-to-Flow Attention for Commuting Flow Prediction. **arXiv preprint, 2024.**

TEACHING EXPERIENCE

Teaching Assistant, The Hong Kong Polytechnic University

Fall 2026	COMP4431 - Artificial Intelligence	Dr. Wai Man Pang (Raymond)
Summer 2024-2026	HTM568 - Smart Tourism and Big Data Analytics	Prof. Chen Jason Zhang
Spring 2026	COMP5112 - Data Structures and Database Systems	Prof. Wen Hua
Fall 2025	COMP5311 - Internet Infrastructure and Protocols	Prof. Donghui Dai
Spring 2025	COMP4442 - Service and Cloud Computing	Prof. Zhaorui Zhang

ACADEMIC SERVICES

Program Committee Member: WSDM 2027.

Conference Reviewer: WWW 2027.

Journal Reviewer: International Journal of Applied Earth Observation and Geoinformation.

Co-reviewer: ACL, VLDB, ICDE, EMNLP, ACM MM, AAAI.

HONORS & AWARDS

Nov. 2018	Second Prize, The Chinese Mathematics Competitions
May 2021	Second-class Scholarship, Chongqing University
Dec. 2020	Advanced Individual of Scientific and Technological Academic Innovation, Chongqing University
Jun. 2020	Third Prize, The 8th TipDM Cup Data Mining Challenge Committee
Jan. 2020; Jan. 2019	Outstanding Student, Chongqing University
Nov. 2018	Third-class Scholarship, Chongqing University

SKILLS

Programming: Python, SQL, R, MATLAB, C++. **English:** IELTS 6.5.